

Digital Notice Board — Screen-by-Screen Developer Specs

(Web Portal + Mobile)

0) Global Definitions (Applies to All Screens)

0.1 Actors / Roles

- ****Super Admin**** (platform owner)
- ****Community Admin**** (management office)
- ****Committee Member**** (approver/publisher depending RBAC)
- ****Staff**** (security/maintenance/admin office; limited)
- ****Resident**** (Owner/Tenant)
- ****Vendor/Contractor**** (limited, optional)
- ****Guest/Temporary**** (optional)

0.2 Announcement Types

- ****General**** (visible to all within community scope)
- ****Targeted**** (subset by targeting rules)
- ****Restricted**** (staff/committee-only or specific roles)
- ****Emergency**** (override rules; forced prominence; special notification behavior)

0.3 Priority Levels (UI badges + ordering)

- ****Normal****
- ****High****
- ****Urgent**** (sticky/pin behavior allowed)
- ****Emergency**** (always top; may use fullscreen takeover)

0.4 Categories (configurable per community)

- Community Updates, Maintenance/Utilities, Events, Security Alerts, Financial/Dues, Admin/Policy, Lost & Found, Religious/Cultural (optional)

0.5 Core Lifecycle / Status Model

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Status	Meaning	Who can move into it
Draft	Not visible to recipients	Creator
Pending Approval	Awaiting approvals	Creator submits
Rejected	Needs edit/resubmit	Approver
Approved	Approved but not published	Approver
Scheduled	Auto-publish at start time	Publisher
Published (Active)	Visible now	Publisher / Scheduler
Expired	Past expiry time	System
Archived	Hidden from default feeds; searchable	Admin/Publisher/System
Deleted (soft)	Recoverable (audit retained)	Super Admin/Admin (policy-based)

0.6 Targeting Model (Rules)

Target rules are evaluated as ****AND of groups****, with ****OR within selections****:

- Community -> Block/Tower -> Floor -> Unit
- Role: Owner/Tenant/Both, Committee, Staff categories
- Individuals
- Custom groups/tags (e.g., Volunteers)
- Optional: age group, gender (only if legally/ethically enabled)

****Emergency****: may ignore targeting and push to “All reachable users”, but still respects “community boundary”.

0.7 Read vs Acknowledgement

- ****Read/Seen****: system marks when a user opens the announcement detail (or scroll threshold; define below).

- ****Acknowledged****: explicit action (“Mark as read” / “I have read & understood”).

- Emergency/Urgent can require acknowledgement.

****Default rule:****

- Read = opening detail (and content loaded) + 2s dwell, or scroll $\geq 40\%$ for long content.

- Acknowledge = button tap + confirmation.

0.8 Notifications (Channels + Quiet Hours)

Channels:

- In-app, Push (mobile), Email (optional), SMS (urgent/emergency), WhatsApp (future)

Quiet Hours:

- Normal/High: respect quiet hours

- Urgent: configurable (respect by default)

- Emergency: ****override quiet hours**** if enabled

0.9 Attachments Rules

- Types: PDF, JPG/PNG, DOC/DOCX, XLS/XLSX, MP4 (admin-controlled), audio optional

- Per-file size limit, total per announcement limit

- Download permission flags:

- View-only (stream/preview)

- Download allowed

- Watermark on download (optional)

0.10 Multi-language

- Per user: preferred language

- Per announcement:

- Single language OR

- Multi-language variants (e.g., EN/UR/AR)

- If missing variant, fallback order: user pref → community default → first available.

1) Screen Inventory (From Wireframes) — Web + Mobile

Web (Portal)

- W1. ****Manage Announcements (Dashboard/List)****
- W2. ****Targeted Announcement (Create + List workspace)****
- W3. ****Scheduling & Expiry (States/Configuration)****
- W4. ****User Feedback (Stats + moderation list)****
- W5. ****Approval Workflow (Queue/List)****
- W6. ****Approval Workflow (Detail + approvals matrix/settings)****
- W7. ****Digital Display Auto-Sync (Integration)****
- W8. ****Security & Audit Logs****

Mobile (App)

- M1. ****Notices Feed (Tabs: General/Targeted/Restricted/Emergency)****
- M2. ****Emergency Broadcast (Full-screen)****
- M3. ****Analytics / Viewership Stats (admin view)****
- M4. ****User Readable Search (Reader experience)****

> Note: Some screens (analytics) appear in mobile wireframes; if analytics is web-only in your product, we keep M3 as “optional/role-gated”.

2) W1 — Web: Manage Announcements (Dashboard/List)

2.1 Screen Overview

- **Name:** Manage Announcements
- **Platform:** Web
- **Primary Purpose:** View, search, filter, and manage all announcements (draft → published → archived)
- **Roles Allowed:** Community Admin, Committee (if granted), Staff (if granted), Super Admin

2.2 Entry Points

- Sidebar: **Notice Board**
- Deep link from notifications (admin alerts: “approval pending”, “scheduled published”)

2.3 UI Components

- Top bar:
 - Community selector (if multi-community)
 - Search input (global within Notice Board)
 - Profile menu
- Page header:
 - Title: “Manage Announcements”
 - Primary CTA: **Create New**
 - Secondary: **Filters**, **Export** (optional), **Analytics** shortcut
- List table/cards (each item shows):
 - Type badge (General/Targeted/Restricted/Emergency)
 - Priority badge (Normal/High/Urgent/Emergency)
 - Title + short excerpt
 - Category
 - Audience summary (“All”, “Block A”, “Owners”, etc.)
 - Status pill (Draft/Pending/Scheduled/Published/Expired/Archived)
 - Publish window (start/expiry)
 - Engagement mini-metrics (Seen %, Ack % if applicable)

- Actions menu (⋮): View, Edit, Duplicate, Archive, Delete, Send Reminder, Force Pin (RBAC)
- Tabs / Quick filters (recommended, matches typical wireframe intent):
 - All / Draft / Pending Approval / Scheduled / Active / Expired / Archived

2.4 Functional Behavior

Search

- Searches across:
 - Title
 - Body (plain-text index of rich content)
 - Category
 - Attachment names
- Debounced search \{(300\text{--}500\)ms
- Permissions: search results only include items user can view.

Filters

- Date range (created/published)
- Category
- Type
- Priority
- Status
- Author
- Audience (block/unit/role)
- “Ack Required” flag

Sorting

- Default: Active first; within active:
 1. Emergency
 2. Urgent (pinned)
 3. High
 4. Normal

- Alternate sort options: Most recent, Most viewed, Most unacknowledged

Row Actions Rules (RBAC)

- ****Edit**** allowed if:
 - Draft OR Pending (if policy allows edit while pending) OR Scheduled (if policy allows)
 - Not allowed if Published unless “edit published” permission exists (then creates a new version)
- ****Duplicate**** always allowed if can create
- ****Archive**** allowed for Published/Expired
- ****Delete**** soft-delete; restricted to Admin/Super Admin
- ****Send Reminder**** only if:
 - Ack required OR important flag
 - Announcement is Active
 - Reminders rate-limited

2.5 Data & Backend Notes

Entities used

- `Announcement`
- `AnnouncementTargetRule`
- `AnnouncementAttachment`
- `AnnouncementCTA`
- `AnnouncementDelivery` (per-channel delivery log)
- `AnnouncementReadReceipt`
- `AnnouncementAcknowledgement`
- `User`, `Unit`, `Block`, `Role`, `GroupTag`
- `AuditLog`

Minimal list API (example)

- `GET /announcements?status=active&search=...&filters...`
- Response includes computed fields:
 - `seen\_count`, `recipient\_count`, `ack\_count`, `unseen\_count`

2.6 QA Acceptance Criteria

- Can filter by each dimension and results are consistent.
- Role-gated actions not visible/clickable.
- Emergency items appear at top regardless of sort (unless explicitly sorted otherwise).
- Search returns items by attachment filename.
- Status transitions reflect schedule and expiry correctly.
- Soft-deleted items disappear from list for non-admin, remain in audit.

3) W2 — Web: Targeted Announcement (Create + Workspace)

3.1 Screen Overview

- **Name:** Create Announcement (Targeted workspace)
- **Platform:** Web
- **Purpose:** Compose and configure a new announcement (type, category, content, attachments, targeting, publish controls)
- **Roles:** Admin, Committee (if can create), Staff (draft-only permission)

3.2 Entry Points

- W1 “Create New”
- Duplicate action from W1 (prefills content)

3.3 UI Components (Mapped to wireframe intent)

Header / Mode

- Announcement Type selector:
 - General / Targeted / Restricted / Emergency
- Category dropdown
- Priority dropdown
- Toggle(s):
 - Pin to top
 - Highlight banner
 - Full-screen takeover (Emergency only)
 - Acknowledgement required (default ON for Emergency; configurable for Urgent)

Form Fields

- Title (required; max length e.g. 120)
- Rich text editor:
 - Bold/Italic/Underline
 - Bullets/Numbering
 - Headings H1–H4

- Links
- Tables
- Inline images
- Emojis (optional; admin toggle)
- CTA buttons builder:
 - Label (required)
 - Type: External URL / Internal route (Pay Now, Register, View Details, Contact Office)
 - Destination + validation
 - Optional: require login

Attachments block

- Upload (drag-drop)
- Preview thumbnails
- Per attachment:
 - Name, type, size
 - Permissions: view-only / downloadable
 - Watermark toggle (optional)
- Storage quota indicator (community plan)

Target Audience Selector (embedded or opens panel)

- Checkboxes:
 - All residents
 - Owners only
 - Tenants only
 - Staff (with category sub-select)
 - Committee
 - Vendors (optional)
 - Custom group(s)
- Location filters:
 - Block/Tower multi-select

- Floors
- Units
- Individuals picker (search user)

Publish controls (can be inline or “Next” step)

- Publish now / Schedule
- Start datetime
- Expiry datetime
- Auto-archive after expiry
- Recurrence (daily/weekly/monthly) with rule builder
- Notification channel toggles (app/push/email/sms/whatsapp)

Preview / Submit

- Preview: Web + Mobile preview tabs
- Buttons:
 - Save Draft
 - Submit for Approval (if workflow enabled)
 - Publish (if user can publish)

3.4 Functional Behavior & Validation

Required validation

- Title required
- Category required
- Content required (non-empty after stripping formatting)
- Targeting:
 - General: target = community default audience
 - Targeted: must select at least one rule (unless “All residents” checked)
 - Restricted: must select restricted roles/groups (cannot be “All”)
 - Emergency: default target = All residents + staff; can restrict only if policy allows
- Scheduling:

- Start <= Expiry
- Expiry optional (if none, remains active until archived manually)
- Attachments:
 - Block disallowed MIME types
 - Enforce file size limits
- CTA:
 - URL valid
 - Internal route must exist and user must have permissions

Versioning rules (important)

- Editing a Published announcement:
 - Option A (recommended): creates ****new version****; old version remains in audit; recipients see latest.
 - Option B: edit-in-place only if permitted; must log in audit with diff.

Emergency behavior rules

- Forces:
 - Priority = Emergency
 - Sticky = ON
 - Full-screen option available
 - Acknowledgement required default ON
 - Notification channels default to Push + In-app (+SMS if enabled)
 - Quiet hours override ON (if enabled in policy)

3.5 Notifications

On Publish:

- Create per-channel delivery jobs.

On Schedule:

- queue job at start datetime.

Reminders:

- Only for Ack-required notices; target “unacknowledged recipients”.

3.6 Data Notes (High-Level Payload)

`POST /announcements`

```
json

{
  "type": "TARGETED",
  "category_id": "maintenance",
  "priority": "URGENT",
  "title": "Water shutdown",
  "body_rich": { "format": "html", "value": "<p>...</p>" },
  "languages": ["en", "ur"],
  "variants": { "en": "...", "ur": "..." },
  "cta_buttons": [{ "label": "Pay Now", "kind": "INTERNAL", "route": "/payments/dues" },
  "attachments": [{ "file_id": "f123", "downloadable": true }],
  "target_rules": { "blocks": ["A"], "roles": ["OWNER"] },
  "ack_required": true,
  "publish": { "mode": "SCHEDULE", "start_at": "...", "end_at": "..." },
  "channels": { "in_app": true, "push": true, "email": false, "sms": false }
}
```

3.7 QA Acceptance Criteria

- Draft saves without targeting (if policy allows) but cannot publish without valid targeting.
- Restricted cannot be visible to residents even if link shared.
- Emergency defaults applied automatically and cannot be downgraded without permission.
- Preview matches content and targeting summary.
- Attachments honor view-only vs download. ---

4) M1 — Mobile: Notices Feed (General/Targeted/Restricted/Emergency)

4.1 Overview

- **Name:** Notices Feed
- **Platform:** Mobile (iOS/Android)
- **Purpose:** Consume announcements; quick acknowledge (“Mark as read”); filter by type

- **Roles:** Residents + Staff + Committee + Admin (feed content differs by RBAC)

4.2 Entry Points

- Bottom tab: **Notices**
- Push notification deep link opens detail but also updates feed

4.3 UI Components

- Header: “Notices” + Search icon + optional filter icon
- Tabs: **General / Targeted / Restricted / Emergency**
- List items show:
 - Icon by category/type
 - Title
 - Timestamp/date
 - Short snippet
 - Badges: Urgent/Emergency
 - “Sender/Author” (optional)
 - Seen/Unseen indicator
- Bottom action (as in wireframe): **Mark as read** (bulk acknowledge for selected or current? see behavior below)

4.4 Behavior

- Tab filters by type but only shows items user has permission to view.
- Tap item → Announcement Detail (screen not explicitly shown in this wireframe set, but **required**; we will spec it under “implicit screens” below).
- “Mark as read” button behavior (define precisely):
 - Option A (recommended): bulk acknowledge **all visible unread** in the current tab after confirmation.
 - Option B: acknowledge **selected** items (requires selection UI).
- We will implement Option A unless you require selection.**
- Pull to refresh
- Infinite scroll pagination

4.5 QA

- Targeted announcements appear only for eligible users.
- Restricted tab not visible to users without any restricted access (optional UX).
- Emergency tab always shows emergency notices, newest first.
- Read/unread changes reflect server state.

5) M2 — Mobile: Emergency Broadcast (Full-screen)

5.1 Overview

- **Name:** Emergency Broadcast
- **Platform:** Mobile
- **Purpose:** High-visibility emergency notice with forced acknowledgement (and optional action buttons)
- **Roles:** All recipients of emergency

5.2 UI Components

- Full-screen card with:
 - Emergency icon/banner
 - Title
 - Body text
 - Action buttons (examples shown in earlier scope: Call security, View details, Directions)
 - Acknowledgement checkbox/button
 - Optional “Confirm / Done”
- Optional: audible alert toggle (policy-based)

5.3 Behavior

- Opens:
 - From push notification
 - From Emergency tab

- Overrides quiet hours if policy enabled
- If acknowledgement required:
 - User must confirm to dismiss (except OS back may still exit; handle by persistent reminder + re-prompt)
- CTA buttons:
 - Call (tel:)
 - Map directions
 - Internal page

5.4 QA

- Emergency broadcast displays even if user not logged in? (policy-dependent; default: requires login)
- Acknowledge recorded once; subsequent opens show “Acknowledged” state.
- If offline, queue acknowledgement and sync later.

6) W3 — Web: Scheduling & Expiry (States)

6.1 Overview

- **Name:** Scheduling & Expiry
- **Purpose:** Configure schedule rules, expiry, recurrence, and automation states
- **Roles:** Admin/Publisher; Committee if granted

6.2 Components

- State selectors: Draft / Scheduled / Active / Expired / Archived
- Date-time pickers
- Recurrence builder (RRULE-like)
- Auto-archive toggle
- Reminder rules:
 - Remind after X hours if unread/unacknowledged
 - Escalation steps (optional)

6.3 Behavior

- Enforces:
 - No recurrence for Emergency unless explicitly enabled
 - No “end_at” earlier than start
- When expired:
 - becomes read-only
 - can be archived automatically if enabled

6.4 QA

- Scheduled item becomes Published exactly at start time (timezone correctness).
- Expiry triggers removal from active feeds.

7) W4 — Web: User Feedback (Stats)

7.1 Overview

- **Name:** User Feedback (Stats)
- **Purpose:** Moderation + engagement reporting (likes/ack/comments/questions/polls if enabled)
- **Roles:** Admin/Moderator; Committee (read-only optional)

7.2 Components

- Filters: category/type/status/date
- Table listing announcements with:
 - Comments count
 - Questions count
 - Likes/ack counts
 - Moderation status (open/locked)
- Drill-down:
 - Comment thread with delete/hide, pin admin reply, lock comments

7.3 Behavior

- Admin can:
 - Disable comments per notice
 - Moderate comments (soft delete + audit)
 - Export feedback report

7.4 QA

- Moderated comments disappear for residents but remain in audit for admins.
- Locked threads prevent new comments immediately.

8) W5 — Web: Approval Workflow (Queue/List)

8.1 Overview

- **Name:** Approval Workflow — Queue
- **Purpose:** Reviewer queue for pending approvals (Maker→Checker→Approver)
- **Roles:** Committee Approver(s), Admin Approver(s), Super Admin

8.2 Components

- Queue tabs: Pending / Approved / Rejected
- Each item shows:
 - Title, type, priority
 - Submitted by + submitted date
 - Required approvals remaining
 - Approve / Reject buttons (if user is current approver)
- Batch approve (optional; not for Emergency unless policy allows)

8.3 Behavior

- Approval decision requires:
 - optional comment on reject
 - audit log entry
- SLA reminders (optional): pending beyond X hours triggers notification.

9) W6 — Web: Approval Workflow (Detail + Matrix)

9.1 Overview

- ****Name:**** Approval Workflow — Detail
- ****Purpose:**** View announcement content + approvals configuration + per-role permissions matrix
- ****Roles:**** Admin/Super Admin; Approvers

9.2 Components

- Left: Announcement preview
- Right: Workflow configuration:
 - Levels (Maker/Checker/Admin)
 - Auto-approval rules for low-risk categories
- Permissions matrix (as in wireframe intent):
 - Roles vs capabilities (Create/Approve/Publish/Archive/View analytics)

9.3 Behavior

- Changing workflow rules affects future submissions (not retroactively unless explicitly “apply to pending”).
- Emergency can bypass workflow only if permission exists; else requires fastest path approval.

10) W7 — Web: Digital Display Auto-Sync

10.1 Overview

- **Name:** Digital Display Auto-Sync
- **Purpose:** Sync selected announcements to physical screens (LED/LCD)
- **Roles:** Admin/Super Admin

10.2 Components

- Device list (screens)
- Content playlist rules:
 - Include categories/types/priorities
 - Rotation timing
 - Ticker mode
 - Emergency override setting
- Preview of layout (optional)

10.3 Behavior

- When Emergency published:
 - instantly overrides playlist if enabled
- Logs delivery to devices

11) W8 — Web: Security & Audit Logs

11.1 Overview

- **Name:** Security & Audit Logs
- **Purpose:** Immutable audit trail for governance/compliance
- **Roles:** Super Admin, Admin (with permission)

11.2 Components

- Filters: actor, action, date range, object type (announcement/comment/attachment)
- Log table includes:
 - Timestamp
 - Actor
 - Action (CREATE/EDIT/PUBLISH/APPROVE/REJECT/DELETE/REMIND)
 - Object ID + title snapshot
 - IP/device metadata (optional)
- Export CSV/PDF (permissioned)

11.3 Behavior

- Edits to published items create audit diff (field-level if feasible).
- Deletions are soft; audit remains.

12) “Implicit but Required” Screens (Not fully shown, but mandatory to not miss features)

These screens must exist for the wireframes to function end-to-end:

1. **Announcement Detail View (Web + Mobile)**

- Full content, attachments preview/download, CTA buttons, comments/Q&A, acknowledgement button, seen state.

2. **Target Audience Picker (Modal/Panel)**

- Blocks/floors/units/roles/groups/individuals; saved presets.

3. **Attachment Viewer (PDF/Image/Video)**

- Permissions enforcement, watermark, streaming.

4. ****Seen/Acknowledgement Report (Admin Drill-down)****

- List of recipients with status + timestamps; export.

5. ****Notification Settings (Community-level + user-level)****

- Quiet hours; channel toggles; emergency overrides.

6. ****Moderation Detail (per announcement)****

- Manage comments, disable thread, admin replies.

7. ****Multi-language Editor (per announcement)****

- Variant management + fallback preview.

> These are included in scope and should be built even if not drawn in the collage.

13) Requirements Traceability (Features → Screens)

Feature Area	Covered In
Types/Categories	W2, M1, W1 filters
Targeting (advanced)	W2 + audience picker
Rich editor + CTAs	W2
Attachments/media	W2 + attachment viewer
Schedule/expiry/recurrence	W3 + W2 publish controls
Priority/pin/fullscreen	W2 + M2 + W1 ordering
Analytics (seen/ack/downloads)	W1 summary + M3 + seen report
Feedback/comments/moderation	W4 + detail screens
Notifications (push/email/sms)	W2 publish controls + settings
Search/history/archive	W1 + M4 + archive filters
Approval workflow	W5 + W6
Digital display integration	W7
RBAC permissions	W6 matrix + global
Multi-language	W2 editor variants + detail
Security/privacy/audit	W8 + attachment permissions
Future AI hooks	optional buttons/flags in W2/detail

Build Notes (Non-Functional)

- Pagination everywhere (lists, logs, recipients)
- Server-side filtering (don't fetch all announcements)
- Cache: feed endpoints should be cacheable per user with short TTL
- Security:
 - Signed URLs for attachments
 - Prevent IDOR (user cannot fetch announcements outside targeting)
 - Audit log is append-only

Screen Spec — Announcement Detail View (Web + Mobile)

1) Screen Overview

- **Screen Name:** Announcement Detail
- **Module:** Digital Notice Board
- **Platforms:** Web (Portal), Mobile (App)
- **Primary Purpose:** Show a single announcement with full content, attachments, CTAs, engagement (ack/like/comment/question), and administrative controls (edit/archive/analytics) based on RBAC.
- **Roles Allowed (View):**
 - **Residents/Staff/Committee/Admin** if the announcement's targeting rules resolve **true** for the user (or user has override permission).
 - **Super Admin/Admin** may have "view all" override (configurable).

2) Entry Points / Navigation

1. From **Notice Feed / List** (Web W1, Mobile M1)
2. From **Push/In-app notification deep link**
 - Example: `/notices/{announcementId}` (web), `app://notices/{announcementId}` (mobile)
3. From **Approval queue** (approver opens detail)
4. From **Analytics** (admin drills into a specific notice)

Deep Link Behavior

- If user is **not authenticated**:
 - Web: redirect to login → return to detail page after auth
 - Mobile: open app login → return to detail after auth
- If user **authenticated but unauthorized** (fails targeting + no override):
 - Show **403-style** message: "You don't have access to this notice."
 - Do **not** leak title/body/attachments.

3) UI Components (Web + Mobile)

3.1 Common Header Block

- **Back** (mobile) / breadcrumb (web)
- **Title**
- Badges:
 - **Type:** General / Targeted / Restricted / Emergency
 - **Priority:** Normal / High / Urgent / Emergency
 - Optional: Category chip
- Meta line:
 - Author (name/role)
 - Published datetime
 - Expiry datetime (if any)
 - Audience summary (e.g., "Block A • Owners")
- Status indicator (admin-only): Draft/Pending/Scheduled/Published/Expired/Archived

3.2 Content Block (Rich Text)

- Rendered rich content:
 - Headings, lists, tables, inline images
 - Hyperlinks (external open confirmation on mobile if required by policy)
- **Multi-language selector** (only shown if multiple variants exist)
- Dropdown: EN / UR / AR etc.
- Default selection follows user preference fallback rules

3.3 Attachments Block

For each attachment:

- Icon + filename + size + type
- Actions (permission-based):
 - **Preview/View** (PDF viewer / image viewer / video player)
 - **Download** (if allowed)

- Optional: Watermark notice (“Downloaded files are watermarked”)

3.4 CTA Buttons Block

- One or more buttons (stacked on mobile, inline on web)
- Supported CTA kinds:
 - **External URL**
 - **Internal route** (e.g., Pay Now → dues page)
 - **Call** (mobile)
 - **Email** (optional)
- Disabled state if destination is not permitted for the user.

3.5 Engagement Block (Resident/Staff)

- **Seen/Read state** indicator (subtle)
- Buttons (feature-flagged per community):
 -  Like
 -  Comment (if enabled)
 -  Ask Question (if enabled; can map to comment type “question”)
 -  Save/Bookmark (optional)
- **Acknowledgement UI** (if `ack\_required = true`)
 - Checkbox “I have read and understood” + **Submit**
 - On mobile wireframe: single **Mark as read** button can function as Acknowledge when ack is required (see logic below)

3.6 Admin Controls (Role-gated)

- Actions menu:
 - Edit
 - Duplicate
 - Archive / Unarchive
 - Delete (soft)
 - Send reminder (to unread/unacknowledged)

- Pin/unpin, highlight toggle
- Analytics shortcut:
 - Seen/Unseen counts
 - Ack completion %
 - Downloads count
 - Comments count

3.7 Comments / Discussion Thread (If Enabled)

- Thread list:
 - Author name + role badge (optional)
 - Timestamp
 - Content
 - Admin reply highlighting
- Composer:
 - Text input
 - Attach image (optional; policy-based)
- Moderation controls (Admin/Moderator):
 - Hide/Delete comment (soft)
 - Lock/unlock comments for this notice
 - Pin admin reply
 - Report/abuse queue (optional)

4) Functional Behavior & Logic

4.1 Access Control (Hard Requirement)

When loading detail:

1. Fetch minimal metadata (or do single fetch with authorization enforced server-side).
2. Server evaluates:

- Community match
- Target rules match OR admin override
- Restricted rules (roles) must be satisfied

3. If unauthorized → return ****403**** without body/attachments.

4.2 Read/Seen Tracking

****Goal:**** accurate “Seen/Unseen” without cheating by list previews.

Seen Event Trigger

Mark ****Seen**** when:

- Detail view is opened and content is successfully rendered ****AND****
- User stays for ≥ 2 seconds ****OR**** scrolls $\geq 40\%$ of content height (whichever first)

Event:

- `POST /announcements/{id}/seen`
- Server stores:
 - `seen\_at`, `device\_type`, `app\_version` (optional)
- Idempotent: multiple posts don’t create duplicates.

Edge Cases

- Offline: queue event and sync later (mobile).
- If announcement is expired but still viewable (archive/search):
 - Seen tracking still allowed (for audit) but may be excluded from “active reach” metrics.

4.3 Acknowledgement Logic (Ack Required)

If `ack\_required = false`:

- “Mark as read” (mobile) toggles ****Seen**** only OR can be hidden; recommended: hide if not needed.

If `ack\_required = true`:

- User must perform explicit action:

- Web: checkbox + submit
- Mobile: “Mark as read” acts as ****Acknowledge**** and opens confirmation if policy requires checkbox
- API:
 - `POST /announcements/{id}/ack`
 - Payload: optional `comment`
- Server stores:
 - `ack\_at`, `comment`, `source`

Reminder Eligibility

- Reminder targets:
 - Users in recipient set where `ack\_required=true` and `ack\_at` IS NULL
 - If ack not required: optionally target unseen users (config)

4.4 Comments / Q&A Behavior

- Commenting allowed only if:
 - Announcement `comments\_enabled=true`
 - User has permission (some communities restrict comments to residents only)
- Moderation:
 - If `moderation\_mode = pre` then comments go to “Pending” until approved
 - If `moderation\_mode = post` comments publish immediately but can be removed

APIs:

- `GET /announcements/{id}/comments?cursor=...`
- `POST /announcements/{id}/comments`
- `PATCH /comments/{commentId}` (hide/delete/restore, admin)
- `POST /comments/{commentId}/report` (optional)

4.5 Attachments Security

- Attachments must use ****signed URLs**** or proxy streaming endpoint.
- Download permission enforced server-side:

- View-only attachments cannot be downloaded via direct URL.
- Watermark:
 - If enabled, watermark applied at download time (PDF/image) or add watermark overlay (video preview) depending on feasibility.

APIs:

- `GET /announcements/{id}/attachments`
- `POST /attachments/{fileId}/signed-url?mode=view|download`

4.6 CTA Execution Rules

- Internal CTA:
 - Validate role permission before navigation
- External URL:
 - Open in system browser (mobile) or new tab (web) with safety flags (`rel="noopener noreferrer"`)
- "Pay Now":
 - If payments module disabled → hide or show disabled state with message.

4.7 Emergency-Specific Behavior

If `type=EMERGENCY` or `priority=EMERGENCY`:

- UI:
 - Prominent banner; optionally full-screen takeover on mobile
- Dismiss rules (if configured):
 - If ack required, block dismiss until ack (within UX constraints of OS back)
 - If not ack required, allow dismiss but keep persistent banner in feed until expiry/archived

5) Data & Backend Notes (High-Level)

5.1 Core Entities

- `Announcement`
- `AnnouncementVariant` (language variants)
- `AnnouncementAttachment`
- `AnnouncementCTA`
- `AnnouncementRecipient` (optional materialized recipient list) OR dynamic targeting evaluation
- `AnnouncementSeen`
- `AnnouncementAck`
- `AnnouncementComment`
- `AuditLog`

5.2 Suggested Endpoints

- `GET /announcements/{id}` → returns full detail (RBAC enforced)
- `POST /announcements/{id}/seen`
- `POST /announcements/{id}/ack`
- `POST /announcements/{id}/like` (optional)
- `GET /announcements/{id}/comments`
- `POST /announcements/{id}/comments`
- `GET /announcements/{id}/analytics` (admin only)

5.3 Metrics Counting

- Seen count = distinct users with `seen\_at`
- Ack count = distinct users with `ack\_at`
- Recipient count:
 - If recipient list materialized at publish time: stable counts
 - If dynamic: compute snapshot at publish and store `recipient\_count\_snapshot` for stable reporting (recommended)

6) Error States

- ****403 Unauthorized:**** show access message; no content leakage
- ****404 Not found:**** “Notice not available”
- ****Expired:**** show read-only banner “Expired on ...”; interactions allowed per policy (comments usually disabled)
- ****Network failure (mobile):**** cached detail (optional) with “offline” indicator; queue seen/ack events

7) QA Acceptance Criteria (Must Pass)

1. ****Targeting enforced****: user outside target cannot view via deep link.
2. ****Seen tracking**** triggers only on detail view with dwell/scroll threshold.
3. ****Ack required**** prevents marking complete without explicit action; status reflects in analytics.
4. ****Attachments****:
 - View-only cannot be downloaded
 - Download URLs expire and cannot be reused
5. ****Multi-language****: correct fallback and manual switch works.
6. ****Comments****:
 - Disabled state blocks posting
 - Moderation rules apply correctly
7. ****Emergency****:
 - Prominent UI always visible
 - Optional full-screen takeover works via push and from feed

Component Spec — Target Audience Selector Panel (Web + Mobile)

1) Component Overview

- **Name:** Audience Selector
- **Used In:** Create/Edit Announcement (W2), also “Duplicate” flow, and optional “Send reminder to...” targeting
- **Platforms:** Web (drawer/modal), Mobile (full-screen stepper)
- **Purpose:** Build targeting rules and compute recipient preview count.

2) Entry Points

- From Create/Edit: **Target Audience** section → “Select audience”
- From Draft validation errors: clicking “Fix audience” opens selector

3) UI Structure

3.1 Header

- Title: “Target Audience”
- Actions:
 - **Save**
 - **Cancel**
 - Optional: **Save as preset**

3.2 Quick Audience Options (Top)

Checkboxes (mutually compatible unless stated):

- ☐ **All community members** (disables other targeting inputs unless “refine” is enabled)
- ☐ Owners
- ☐ Tenants
- ☐ Committee members

- ☐ Staff (expand → categories: security/maintenance/admin)
- ☐ Vendors/Contractors (optional feature flag)
- ☐ Custom Groups/Tags (multi-select)

3.3 Location Targeting

Cascading multi-select:

- Blocks/Towers (multi)
- Floors (multi; filtered by selected blocks)
- Units (multi; filtered by selected floors)

Behavior:

- Selecting Units auto-selects parent Floor/Block (display as derived).

3.4 Individual Users

- Search input (name/phone/unit)
- Multi-select users

Rules:

- Individual selection is ****OR**** with other targeting rules only if “Include individuals” is enabled; otherwise it can act as override list.

Recommended:

- Implement as ****union**** (adds recipients) with a toggle:
 - (•) Add to recipients
 - () Only these individuals (confidential mode) — permission-gated

3.5 Advanced Targeting (Feature-flagged)

- Gender-based (if allowed)
- Age group (optional)
- RBAC tags (e.g., “Event Volunteers”)
- “Exclude” rules (optional but powerful):
 - Exclude blocks/units/users
 - Exclude role (e.g., exclude vendors)

3.6 Recipient Preview Panel

- Shows:
 - Estimated recipient count
 - Breakdown (Owners/Tenants/Staff)
- Refresh behavior:
 - Live update after each change (debounced) OR manual “Preview recipients”
- If count exceeds limit thresholds:
 - show warning (e.g., SMS cost, push rate limits)

4) Targeting Logic (Deterministic)

Represent targeting as:

- `include` rules (AND across dimensions; OR within same dimension)
- optional `exclude` rules
- optional `mode`:
 - `DYNAMIC` (evaluate at view time)
 - `SNAPSHOT` (materialize recipients at publish time) — recommended for analytics stability

Validation Rules

- Targeted type requires at least one include rule ****unless**** “All community members” checked.
- Restricted type cannot use “All community members” unless restricted to staff/committee by role.
- Emergency defaults to All (unless policy allows restricting emergencies).

5) Backend / API Notes

5.1 Preview Endpoint

- `POST /announcements/targeting/preview`

json

```
{
  "community_id": "c1",
  "include": {
    "blocks": ["A", "B"],
    "floors": [2, 3],
    "roles": ["OWNER", "TENANT"],
    "staff_categories": ["SECURITY"],
    "groups": ["VOLUNTEERS"],
    "users": ["u1", "u2"]
  },
  "exclude": {
    "units": ["A-301"]
  },
  "mode": "SNAPSHOT"
}
```

Response:

json

```
{
  "estimated_count": 182,
  "breakdown": { "owners": 90, "tenants": 80, "staff": 12 },
  "warnings": ["SMS to 182 recipients may incur cost"]
}
```

5.2 Persisting Rules on Announcement

- Store the rule JSON + a normalized set for querying.
- If `SNAPSHOT`, store recipient IDs at publish time (or store hashed membership set) depending on scale.

6) QA Acceptance Criteria

1. Cascading selects filter correctly (block → floors → units).
2. “All community members” disables conflicting inputs (or clearly applies precedence).
3. Preview counts match final published recipient set (within agreed rules).
4. Restricted announcements cannot be targeted to residents unless permission explicitly granted.
5. Exclusion rules remove recipients correctly.
6. Individual-only confidential targeting is permission-gated and not discoverable by unauthorized roles.

Immediate Next Deliverable (continuing the sequence)

- ****Screen Spec:**** Attachments Viewer (PDF/Image/Video) + Download Permissions + Watermarking
- ****Screen Spec:**** Seen/Acknowledgement Report (Admin drill-down) + Export (PDF/Excel) + Reminder sending rules

Screen Spec — Attachments Viewer (PDF / Image / Video) + Download Permissions + Watermarking

1) Screen Overview

- **Screen Name:** Attachment Viewer
 - **Module:** Digital Notice Board
 - **Platforms:** Web + Mobile
 - **Primary Purpose:** Securely preview and (optionally) download announcement attachments with strict permission enforcement and optional watermarking.
 - **Roles Allowed:** Any user who can view the parent announcement **and** has attachment-level permission.
-

2) Entry Points / Navigation

1. From **Announcement Detail** → tap/click an attachment item
2. From **Search results** (if attachment search enabled) → opens parent announcement then attachment
3. From **Admin analytics** (download audit) → opens attachment (admin-only)

3) Attachment Types & Supported View Modes

Type	Preview Mode	Download Support	Notes
PDF	Embedded viewer	Optional	Watermark supported (server-side best)
Image (JPG/PNG)	Image viewer + zoom	Optional	Watermark supported (server-side or overlay for preview)
Video (MP4)	Streaming player	Optional	Prefer streaming; watermark overlay for preview may be used
DOC/XLS	Preview (optional) or "Open with"	Optional	If preview not supported, show download/open only
Audio (optional)	Player	Optional	Admin-controlled

4) UI Components

4.1 Viewer Header

- Back button (mobile) / close (web modal) / breadcrumb (web page)
- Filename + file size
- Attachment badges:
 - File type (PDF/IMG/VID)
 - Permission badge: ****View-only**** / ****Downloadable****
 - Watermark badge: ****Watermarked**** (if enabled)

4.2 Viewer Body

PDF

- Page thumbnails (web optional)
- Page navigation, zoom, search within PDF (if viewer supports)
- “Open in new tab” (web optional, permission-gated)

Image

- Pinch/zoom (mobile), zoom controls (web)
- Swipe between images (if multiple attachments and user chooses “View all media”)

Video

- Play/pause, scrub, full-screen
- Streaming quality selection (optional)

4.3 Actions Footer / Menu

- ****Download**** (only if allowed)
- ****Share**** (mobile; only if policy allows sharing downloadable content)
- ****Report issue**** (optional)

- **Copy link** (web; only if link is permission-safe—generally avoid)

5) Permissions & Security (Hard Requirements)

5.1 Access Check (Always Server-Enforced)

A user may access an attachment only if:

1. User is authorized to view the **parent announcement** (targeting + RBAC), and
2. Attachment permission allows the requested mode:
 - **Preview** allowed if ``can_preview=true``
 - **Download** allowed if ``can_download=true``

> UI must not rely on hiding buttons only; backend must enforce both preview and download.

5.2 URL Strategy

Use **short-lived signed URLs** (or proxy endpoints), with distinct scopes:

- ``view`` scope: streaming/preview only
- ``download`` scope: forces download, includes watermark if enabled

Signed URL constraints:

- TTL: e.g., 60–300 seconds
- Bound to ``user_id``, ``attachment_id``, ``scope``, optionally ``device_id``
- One-time token optional for high sensitivity

5.3 Preventing IDOR

- Attachment IDs must not be directly fetchable without authorization checks.
- Any storage object keys must be non-guessable.

6) Watermarking Rules

6.1 When to Apply

Watermark can be enabled at:

- Community level (default on/off)
- Announcement level override
- Attachment level override

Apply watermark on:

- ****Download**** (recommended, enforceable)
- Preview watermark optional (overlay) to discourage screenshots, but not foolproof.

6.2 Watermark Content (Configurable Template)

Include minimal but useful attribution:

- Community name
- User display name or unit (if allowed)
- Timestamp
- Optional: unique trace code (hash)

Example watermark string:

- `GreenView • Unit A-203 • 2026-01-28 14:22 • WID: 8F3K2`

6.3 PDF Watermark Implementation (Recommended)

- Server-side watermark stamping per download request.
- Flatten layer to reduce removal.
- If stamping fails, fallback to non-watermarked download only if policy allows; otherwise block with error.

6.4 Image Watermark

- Server-side watermark on download (preferred)
- Position: diagonal or bottom-right; opacity 15–30%
- Preserve EXIF policy (usually strip EXIF for privacy)

6.5 Video Watermark (Practical Approach)

Options:

- ****Overlay during streaming**** (UI layer watermark) for preview
- ****Burn-in watermark on download**** is expensive; enable only for high-tier plans or skip and log downloads heavily

7) Audit & Analytics Events

Track:

- `attachment\_viewed` (preview opened)
- `attachment\_downloaded` (download started/completed)

Store:

- user_id, announcement_id, attachment_id
- timestamp
- device/app metadata
- IP (web), optional

8.1 Fetch Attachment Metadata

- `GET /announcements/{announcementId}/attachments` Returns list with permission flags:

json



```
[
  {
    "attachment_id": "att_1",
    "name": "maintenance_notice.pdf",
    "mime": "application/pdf",
    "size_bytes": 382001,
    "can_preview": true,
    "can_download": false,
    "watermark_on_download": true
  }
]
```

8.2 Get Signed URL

- `POST /attachments/{attachmentId}/signed-url`

json



```
{ "scope": "view", "announcement_id": "ann_1" }
```

Response:

json



```
{ "url": "https://...", "expires_in": 120 }
```

8.3 Download Endpoint (Alternative)

- `GET /attachments/{attachmentId}/download?announcement\_id=ann\_1`

Server:

- checks auth + permission
- applies watermark if enabled
- streams file

9) Error States

- ****403****: “You don’t have permission to access this file.”
- ****404****: “File not found.”
- ****410 Gone****: attachment expired/auto-deleted
- ****Watermark failure****: show reason + retry; block download if watermark required.

10) QA Acceptance Criteria (Must Pass)

1. View-only attachments: preview works; ****download is blocked**** server-side.
2. Signed URLs expire and cannot be reused after TTL.
3. Watermarked downloads contain correct user/community attribution.
4. Unauthorized user cannot access attachment via direct URL guessing.
5. Video streams without full file download (where supported).
6. Audit logs record view/download events accurately.

Screen Spec — Seen/Acknowledgement Report (Admin Drill-Down) + Export (PDF/Excel) + Reminder Rules

1) Screen Overview

- **Screen Name:** Notice Reach Report (Seen/Ack Drill-Down)
- **Module:** Digital Notice Board → Analytics
- **Platform:** Web (primary). Mobile admin view optional.
- **Primary Purpose:** Show recipient list and compliance status for a specific announcement; export reports; send reminders to unread/unacknowledged recipients.
- **Roles Allowed:** Admin, Committee (if granted analytics permission), Super Admin

2) Entry Points

1. From **Manage Announcements (W1)** → click metrics (Seen/Ack) or “Analytics”
2. From **Announcement Detail (admin controls)** → “View Analytics”
3. From **Analytics dashboard** → select an announcement

3) UI Components

3.1 Header Summary

- Announcement title + badges (type/priority)
- Audience summary + recipient count (snapshot)
- Key metrics tiles:
 - Total recipients
 - Seen count + %
 - Unseen count + %
 - Acknowledged count + % (if ack required)
 - Unacknowledged count
 - Attachment downloads count
 - Comments/questions count (optional)

3.2 Filters & Controls

- Search (by name, unit, phone masked, role)
- Filters:
 - Status: Seen/Unseen/Acked/Unacked
 - Role: Owner/Tenant/Staff categories/Committee
 - Location: Block/Floor/Unit
 - Time windows: seen date range, ack date range
- Sorting:
 - Name
 - Unit
 - Seen time (recent/oldest)
 - Ack time
- Actions:
 - ****Send Reminder****
 - ****Export PDF****
 - ****Export Excel****
 - Optional: ****Copy summary**** (text)

3.3 Recipient Table

Columns (configurable by privacy policy):

- Resident/Staff Name
- Unit / Block
- Role
- Delivery status (optional per channel: Push delivered, Email sent, SMS sent)
- Seen status + `seen\_at`
- Ack status + `ack\_at` (if applicable)
- Last reminder sent at
- Notes (optional: ack comment)

Row actions (admin-only):

- View user profile (if permitted)
- Send reminder to this user (rate-limited)
- Mark as exempt (optional governance feature)

4) Core Logic & Computation

4.1 Recipient Set (Stability Requirement)

For compliance metrics, recipient set should be **stable**:

- Recommended: at publish time, store `recipient\_count\_snapshot` and (for drill-down) a `recipient\_snapshot` list or computable snapshot rule version.
- If dynamic targeting is used, drill-down must clearly label: "Recipient list is dynamic; counts may change." (Not recommended for compliance.)

4.2 Definitions

- **Seen**: user has `seen\_at` recorded (per announcement)
- **Acknowledged**: user has `ack\_at` recorded
- **Compliance Needed**: only if `ack\_required=true`

4.3 Timezones

- Display times in **community timezone** by default; allow admin to switch to UTC (optional).

5) Reminder Sending Rules (No Surprises)

5.1 Eligibility

A reminder can be sent when:

- Announcement status is **Published (Active)** (optionally allow for Expired if compliance still required)
- User is in recipient snapshot
- User is:
 - **Unacknowledged** if `ack\_required=true`, else
 - **Unseen** if reminders for seen are enabled
- Not rate-limited (see below)

5.2 Rate Limiting (Recommended Defaults)

- Per announcement per user:
 - Max 1 reminder per 24 hours
 - Max 3 reminders total (configurable)
- Per announcement global:
 - Batch reminder max recipients per action (e.g., 500)
- Channel fallback:
 - Try Push → In-app → Email → SMS (based on enabled channels and urgency)

5.3 Channels & Templates

- Admin selects channels (if permitted):
 - In-app (always safe)
 - Push
 - Email
 - SMS (cost + policy gated)
- Reminder message template:
 - “Reminder: Please read and acknowledge: {title}”
- Include deep link to announcement.

5.4 Auditing

Every reminder batch creates an audit record:

- who triggered it

- filters used (e.g., “unacknowledged only”)
- recipient count
- channels used
- job id + delivery results summary

6) Export (PDF/Excel)

6.1 Export Scope

Exports must reflect current filter state:

- If filtered to “Unacknowledged”, export only those rows (and indicate filters on export header).

6.2 PDF Export

- Summary header (announcement title, published date, audience, metrics)
- Paginated table
- Timestamp + exported by
- Optional watermark “Confidential”

6.3 Excel Export (XLSX)

Columns:

- Name (or masked if policy)
- Unit
- Role
- Seen (Y/N), seen_at
- Ack (Y/N), ack_at
- Delivery status (optional)
- Reminders sent count, last reminder at

Privacy:

- Mask phone/email unless super-admin permission.

7) Backend/API Notes (High-Level)

7.1 Analytics Summary

- `GET /announcements/{id}/analytics/summary` Returns stable counts + flags:

json

```
{
  "recipient_count": 180,
  "seen_count": 145,
  "ack_required": true,
  "ack_count": 120,
  "download_count": 32
}
```

7.2 Recipient Drill-Down

- `GET /announcements/{id}/analytics/recipients?status=unack&block=A&cursor=...`

Return:

- rows + cursor
- include ``last_reminder_at``, ``reminder_count``

7.3 Send Reminder (Batch)

7.3 Send Reminder (Batch)

- `POST /announcements/{id}/reminders`

json

```
{
  "target": "UNACKNOWLEDGED",
  "filters": { "block": ["A"], "role": ["OWNER"] },
  "channels": { "in_app": true, "push": true, "email": false, "sms": false },
  "message_override": null
}
```

Response:

- job accepted + estimated recipients + warnings (cost/rate limits)

7.4 Export Jobs (Async Recommended)

- `POST /announcements/{id}/analytics/export`

json

```
{ "format": "XLSX", "filters": {...} }
```

- `GET /exports/{exportId}` → download when ready

8) Error & Edge Cases

- Announcement deleted/archived: exports still possible if user has permission (audit requirement).
- Recipient snapshot missing: block export/drill-down with admin message "Recipient snapshot not available; republish required for compliance tracking."
- Large communities: require async export jobs and paginated recipient endpoints.

9) QA Acceptance Criteria (Must Pass)

1. Counts on summary match drill-down totals under “All” (within same snapshot).
 2. Filters apply correctly and persist into exports.
 3. Reminder:
 - sends only to eligible users
 - respects rate limits (verify repeated sends blocked)
 4. Permissioning:
 - only authorized roles can view recipient identities and export data
 - masked fields obey policy
 5. Export files:
 - include correct filters metadata
 - contain correct timestamps and “exported by”
 6. Audit log entries created for exports and reminder actions.
-

****YES.****

Next Screens (to complete the attached wireframe set)

I’m continuing with the remaining screens that weren’t fully specified yet, starting ****now****:

- ****M3 — Mobile: Viewership Stats (Analytics)****
- ****M4 — Mobile: User Readable Search****
- ****W7 — Web: Digital Display Auto-Sync (expanded implementation spec)****
- ****W4 — Web: User Feedback (moderation detail + controls, expanded)****
- ****W9 — Web/Mobile: Notification Controls (per notice + community defaults)****
- ****W10 — Web: Archive & Search (advanced history)****

Below are the ****next two**** (M3 + M4) in full dev-spec format.

Screen Spec — M3 Mobile: Viewership Stats (Analytics)

1) Overview

- **Screen Name:** Viewership Stats
- **Module:** Notice Board → Analytics
- **Platform:** Mobile
- **Purpose:** Show announcement performance metrics and allow quick drill-down for admins/committee.
- **Roles Allowed:** Admin, Committee (if `can\_view\_analytics=true`), Super Admin
 - Non-analytics roles must not see this screen or must see “Not authorized”.

2) Entry Points

- From Mobile Notices (M1) → overflow (...) on an announcement → **Analytics**
- Deep link from admin notification (“Engagement report ready”)

3) UI Components

- Header: “Viewership Stats” + back
- Announcement picker (dropdown or “Current notice” card)
- Metrics cards:
 - Total Recipients
 - Seen (count + %)
 - Unseen
 - Acknowledged (if ack required)
 - Unacknowledged (if ack required)
 - Downloads (optional)
 - Comments / Questions (optional)
- Drill-down buttons:
 - **View Seen List**
 - **View Unseen List**
 - **View Unacknowledged List** (only if ack required)
- Filters (optional, but recommended for parity with web):
 - Block/Unit, Role, Date range

4) Functional Behavior

- Metrics are based on ****recipient snapshot**** at publish time (preferred).
- Drill-down opens a paginated list:
 - Name (or masked per privacy policy)
 - Unit
 - Status badges (Seen/Ack)
 - Timestamp (seen_at / ack_at)
- If announcement is ****Draft/Pending****:
 - Show message: "Analytics available after publish."
- If announcement is ****Expired/Archived****:
 - Analytics remains viewable (audit/compliance).

5) Data/API Notes

- `GET /announcements/{id}/analytics/summary`
- `GET /announcements/{id}/analytics/recipients?status=seen|unseen|ack|unack&cursor=...`

6) QA Acceptance Criteria

- Unauthorized roles cannot access analytics via deep link.
- Seen/Ack counts match server summary.
- Drill-down pagination stable and consistent with filters.
- Ack-related fields hidden when `ack\_required=false`.

Screen Spec — M4 Mobile: User Readable Search (Notices Search)

1) Overview

- ****Screen Name:**** Notice Search
- ****Module:**** Notice Board
- ****Platform:**** Mobile
- ****Purpose:**** Let users search notices (including archived if permitted) by keyword, category, and date.
- ****Roles Allowed:**** All authenticated users (results constrained by targeting/RBAC).

2) Entry Points

- Search icon on Notices feed (M1)
- Optional global app search → “Notices” vertical

3) UI Components

- Search bar (sticky)
- Recent searches (optional)
- Filters:
 - Date range (From–To)
 - Category
 - Priority (Normal/High/Urgent/Emergency)
 - Status: Active / Expired / Archived (Archived visibility policy-based)
- Results list items:
 - Title, snippet with highlights
 - Badges (priority/type)
 - Timestamp
 - Attachment icon if any
- Empty state with suggestions (“Try fewer words”, “Check spelling”)

4) Functional Behavior

- Search scope includes:
 - Title

- Body text (indexed from rich content)
- Attachment filenames (if enabled)
- Permission enforcement:
 - Results only include notices user can access via targeting rules.
- Tapping result opens **Announcement Detail**.
- Debounced input `\(300\text{--}500\)ms`; server-side search.

5) Data/API Notes

- ``GET /announcements/search?q=...&category=...&status=...&from=...&to=...&cursor=...``
- Server must apply RBAC/targeting filters before returning.

6) QA Acceptance Criteria

- User cannot find restricted notices via search.
- Search returns matches from attachment filenames when enabled.
- Filters correctly narrow results; pagination works.
- Opening a result updates Seen tracking only after detail render threshold.

Screen Spec — Analytics Dashboard (Web)

1) Screen Overview

- **Screen Name:** Analytics Dashboard
- **Module:** Digital Notice Board → Analytics
- **Platform:** Web (Portal)
- **Primary Purpose:** Provide a community-wide analytics view across announcements: reach, compliance (ack), engagement, trends, and drill-down to per-notice reports.
- **Roles Allowed:** Super Admin, Community Admin, Committee (if `can\_view\_analytics=true`)
- **RBAC Notes:**
 - Users without analytics permission must not see the menu item and must receive **403** if deep-linked.

2) Entry Points / Navigation

1. Sidebar: **Notice Board → Analytics**
2. From **Manage Announcements (W1)**: “Analytics” button
3. Deep link from export job completion notification (optional)

3) UI Components (Layout)

3.1 Header Bar

- Page title: **Notice Board Analytics**
- Community selector (if multi-community)
- Date range picker (default: last 30 days)
- Quick chips:
 - Last 7 days / 30 days / 90 days / Custom
- Global filters button (opens filter panel)
- Actions:

- **Export Dashboard** (PDF)
- **Export Data** (Excel/CSV)

3.2 KPI Tiles (Top Row)

Show aggregated metrics for selected date range and filters:

- **Total Announcements Published**
- **Total Recipients (snapshot-based sum)** (see calculation rules)
- **Avg Seen Rate** $((\frac{\text{sum seen}}{\text{sum recipients}}))$
- **Avg Ack Rate** (only for ack-required announcements)
- **Total Unacknowledged** (ack-required only)
- **Total Attachment Downloads**
- **Total Comments / Questions** (if enabled)
- **Emergency Count** + avg time-to-ack (optional)

> Each KPI tile supports click → opens relevant filtered listing (see section 3.5).

3.3 Charts Section

Minimum set (matches your module scope and what dev teams need):

1. **Reach Trend (Seen %) Over Time**
 - Line chart: daily/weekly
 - Series: Seen %, Ack % (toggle)
2. **Announcements by Category**
 - Bar chart: category → count + avg seen %
3. **Announcements by Priority**
 - Donut/bar: Normal/High/Urgent/Emergency
4. **Engagement Breakdown**

- Likes, Comments, Questions, Downloads (stacked bar or tiles)

5. ****Top Performing Notices****

- Table snippet with “View report” link

3.4 Filter Panel (Right Drawer)

Filters apply to all tiles/charts/tables:

- Category (multi)
- Type (General/Targeted/Restricted/Emergency)
- Priority (multi)
- Status (Published/Expired/Archived; default: Published within date range)
- Author (admin/committee/staff)
- Audience dimensions (optional if you want segmentation):
 - Block/Tower, Floor, Role (Owner/Tenant/Staff category)
- “Ack required only” toggle
- “Include Restricted” toggle (only if role has permission)

3.5 Announcement Analytics Table (Bottom)

A paginated table of announcements matching filters:

Columns:

- Title
- Category
- Type
- Priority
- Published at
- Recipient count (snapshot)
- Seen % (and seen count)
- Ack % (if applicable)
- Downloads
- Comments

- Actions:
 - ****View Report**** (opens Seen/Ack Drill-down screen)
 - ****Open Notice**** (opens Announcement Detail)
 - ****Export Report**** (PDF/Excel) (permission-gated)

4) Functional Behavior & Logic

4.1 Filter + Date Range Behavior

- All dashboard widgets update when:
 - Date range changes
 - Filters change
- Debounce filter updates $\backslash(300\backslash\text{-}500\backslash)$ ms to prevent API spam.
- Persist selected filters in URL query params for shareable deep links (permission-safe), e.g.:
 - ``/notice-analytics?from=...&to=...&category=maintenance&type=targeted``

4.2 Metrics Calculation Rules (Consistency Requirement)

To avoid “numbers don’t match” issues, define this clearly:

- ****Announcement Recipient Count:**** use ****recipient snapshot at publish time**** (``recipient_count_snapshot``).
- ****Total Recipients (dashboard):**** sum of ``recipient_count_snapshot`` across announcements in scope.
 - Note: this can exceed unique user count because users can receive multiple notices (expected).
- ****Seen Rate (dashboard):****

- **Seen Rate (dashboard):**

$$\text{SeenRate} = \frac{\sum \text{seen_count}}{\sum \text{recipient_count}}$$

- ****Ack Rate:**** same pattern but only where ``ack_required=true``.
- If snapshot not available for an old notice:

- Either exclude from aggregated rates and show “Data completeness: 96%”
- Or compute dynamic recipients but label clearly (not recommended)

4.3 Drill-Down Behavior

- Clicking a row action ****View Report**** opens:
 - ****Seen/Acknowledgement Report (Admin Drill-down)**** with the announcement ID and filter context.
- Clicking KPI tiles:
 - Opens the table with an auto-applied filter, e.g.:
 - “Total Unacknowledged” → table filtered to `ack\_required=true` and sorted by highest unack count.

4.4 Export Behavior

Export Dashboard (PDF)

- Includes:
 - Selected date range + active filters summary
 - KPI tiles
 - Charts (as images)
 - Top table (first N rows) or full table summary (configurable)
- Generated asynchronously for large charts/communities.

Export Data (Excel/CSV)

- Exports the announcement analytics table dataset for the current filter scope:
 - One row per announcement with metrics columns
- For privacy:
 - Does ****not**** include recipient-level names (that’s only in drill-down with extra permission).

4.5 Permissions & Data Privacy

- Restricted announcements:
 - Only included if user has `can\_view\_restricted\_analytics=true`
 - Otherwise excluded entirely (not just hidden titles)
- Masking:

- No personal user data shown on this dashboard.
- Recipient drill-down handles identity exposure controls.

5) Backend / API Notes (High-Level)

5.1 Dashboard Summary Endpoint

- `GET /noticeboard/analytics/summary?from=...&to=...&filters=...` Returns KPIs + completeness info:

json

```
{
  "kpis": {
    "announcements": 42,
    "total_recipients": 8120,
    "total_seen": 6500,
    "total_ack_required": 18,
    "total_ack": 4200,
    "total_downloads": 380,
    "total_comments": 96
  },
  "data_completeness": { "snapshot_coverage": 0.99 }
}
```

5.2 Charts Endpoints

- `GET /noticeboard/analytics/trends?...`
- `GET /noticeboard/analytics/by-category?...`
- `GET /noticeboard/analytics/by-priority?...`

5.3 Table Endpoint

- `GET /noticeboard/analytics/announcements?from=...&to=...&filters...&cursor=...`
Returns paginated rows.

5.4 Export Endpoint (Async)

- `POST /noticeboard/analytics/export` Payload:

json

```
{ "format": "PDF", "from": "...", "to": "...", "filters": {...} }
```

Returns `export_id`, then:

- `GET /exports/{export_id}`

6) Error States

- ****403:**** not authorized
- ****Empty state:**** “No announcements match your filters”
- ****Partial data:**** show banner “Some older notices lack snapshot data; metrics shown for available data only”
- ****Export failed:**** show retry + audit entry

7) QA Acceptance Criteria (Must Pass)

1. Dashboard is inaccessible without analytics permission (UI + API).
2. Changing date range/filters updates KPIs, charts, and table consistently.
3. Aggregated rates match the sum-based formulas and align with per-notice drill-down totals.
4. Restricted announcements are excluded entirely for unauthorized roles (no leakage in counts or charts).
5. Exports reflect active filters and include filter summary text.
6. Pagination and sorting of the table are stable under filtering.
7. Large datasets do not time out (async exports, server-side aggregation, caching where appropriate).

Which **database** should I target for the schema: **PostgreSQL** or **MySQL**? *(If you don't specify, I'll proceed with **PostgreSQL** defaults.)*

Yes — I can perform all three

- **OpenAPI spec** (developer-ready contract)
- **DB schema** (tables, keys, indexes)
- **Backlog** (Epics → Stories → Tasks mapped to the screens)

DB Schema (PostgreSQL) — Core Tables

> Notes:

> - IDs shown as `uuid`.

> - Multi-tenant = `community\_id` on all business tables.

> - "Recipient snapshot" is included to make analytics stable.

1.1 Reference / Tenant Structure

`communities`

- `id` uuid pk`
- `name` text not null`
- `timezone` text not null default 'UTC`
- `default\_language` text not null default 'en`
- `created\_at` timestampz not null default now()`

`users`

- `id` uuid pk`
- `community\_id` uuid not null fk -> communities(id)`
- `full\_name` text not null`
- `email` text null`
- `phone` text null`
- `preferred\_language` text null`
- `status` text not null default 'ACTIVE'` *(ACTIVE/SUSPENDED)*
- `created\_at` timestampz not null default now()`

****Indexes****

- `(community\_id, email)` unique where email not null
- `(community\_id, phone)` unique where phone not null

`roles`

- `id` uuid pk`
- `community\_id` uuid not null fk`

- `code` text not null` *(SUPER_ADMIN, COMMUNITY_ADMIN, COMMITTEE, STAFF_SECURITY, STAFF_MAINT, RESIDENT_OWNER, RESIDENT_TENANT, etc.)*

- `name` text not null`

****Indexes****

- unique `(community\_id, code)`

`permissions`

- `id` uuid pk`

- `code` text not null unique` *(notice.create, notice.publish, notice.analytics.view, notice.approval.approve, etc.)*

- `description` text`

`role\_permissions`

- `role\_id` uuid fk`

- `permission\_id` uuid fk`

- pk `(role\_id, permission\_id)`

`user\_roles`

- `user\_id` uuid fk`

- `role\_id` uuid fk`

- pk `(user\_id, role\_id)`

1.2 Location / Audience Model

`blocks`

- `id` uuid pk`

- `community\_id` uuid fk`

- `name` text not null`

`units`

- `id` uuid pk`
 - `community\_id` uuid fk`
 - `block\_id` uuid fk -> blocks(id)`
 - `floor` int null`
 - `unit\_no` text not null` *(e.g., A-203)*
- **Indexes****
- unique `(community\_id, unit\_no)`

`user\_unit\_links`

- `user\_id` uuid fk -> users(id)`
- `unit\_id` uuid fk -> units(id)`
- `occupancy\_type` text not null` *(OWNER/TENANT)*
- pk `(user\_id, unit\_id, occupancy\_type)`

`groups`

- `id` uuid pk`
- `community\_id` uuid fk`
- `name` text not null`
- `type` text not null default 'CUSTOM'` *(CUSTOM/TAG/SYSTEM)*
- `created\_at` timestampz default now()`

`group\_members`

- `group\_id` uuid fk -> groups(id)`
- `user\_id` uuid fk -> users(id)`
- pk `(group\_id, user\_id)`

`announcement\_categories`

- `id` uuid pk`
- `community\_id` uuid fk`
- `name` text not null`
- `is\_active` bool not null default true`

`announcements`

- `id` uuid pk`
- `community\_id` uuid not null fk`
- `type` text not null` *(GENERAL/TARGETED/RESTRICTED/EMERGENCY)*
- `priority` text not null` *(NORMAL/HIGH/URGENT/EMERGENCY)*
- `category\_id` uuid fk -> announcement_categories(id)`
- `title` text not null`
- `body\_format` text not null` *(HTML/MD)*
- `body` text not null` *(default language body or canonical)*
- `ack\_required` bool not null default false`
- `comments\_enabled` bool not null default true`
- `pinned` bool not null default false`
- `highlight\_banner` bool not null default false`
- `fullscreen\_takeover` bool not null default false`
- `status` text not null`
(DRAFT/PENDING_APPROVAL/APPROVED/SCHEDULED/PUBLISHED/EXPIRED/ARCHIVED/DELETED)
- `starts\_at` timestampz null`
- `ends\_at` timestampz null`
- `auto\_archive` bool not null default true`
- `created\_by` uuid not null fk -> users(id)`
- `published\_by` uuid null fk -> users(id)`
- `created\_at` timestampz not null default now()`
- `updated\_at` timestampz not null default now()`

- `deleted\_at timestamptz null`

****Indexes****

- `(community\_id, status, starts\_at)`

- `(community\_id, created\_at desc)`

- Full-text index (Postgres): `to\_tsvector('simple', title || ' ' || body)`

`announcement\_variants`

- `id uuid pk`

- `announcement\_id uuid not null fk -> announcements(id)`

- `language text not null` *(en/ur/ar)*

- `title text not null`

- `body\_format text not null`

- `body text not null`

****Indexes****

- unique `(announcement\_id, language)`

`announcement\_ctas`

- `id uuid pk`

- `announcement\_id uuid fk`

- `label text not null`

- `kind text not null` *(EXTERNAL_URL/INTERNAL_ROUTE/CALL/EMAIL)*

- `target text not null` *(url/route/phone/email)*

- `sort\_order int not null default 0`

`announcement\_attachments`

- `id uuid pk`

- `announcement\_id uuid not null fk`

- `file\_name text not null`

- `mime\_type text not null`

- `size\_bytes bigint not null`

- `storage\_key` text not null `\*(object storage key)\*`
- `can\_preview` bool not null default true`
- `can\_download` bool not null default true`
- `watermark\_on\_download` bool not null default false`
- `created\_at` timestampz default now()`

****Indexes****

- `(announcement\_id)`

1.4 Targeting Rules + Recipient Snapshot

`announcement\_target\_rules`

- `id` uuid pk`
- `announcement\_id` uuid not null fk`
- `rules` jsonb not null`
- `\*(stores include/exclude blocks/units/roles/groups/users/staff\_categories etc.)\*`
- `mode` text not null default 'SNAPSHOT' `\*(SNAPSHOT/DYNAMIC)\*`
- `created\_at` timestampz default now()`

`announcement\_recipients`

(materialized snapshot; one row per recipient at publish time)

- `announcement\_id` uuid fk`
- `user\_id` uuid fk`
- `delivery\_eligible` bool not null default true`
- `created\_at` timestampz default now()`
- pk `(announcement\_id, user\_id)`

****Indexes****

- `(announcement\_id, user\_id)`

`announcement\_metrics\_snapshots`

- `announcement\_id` uuid pk fk`
- `recipient\_count` int not null`
- `created\_at` timestamptz default now()`

1.5 Read / Ack / Engagement

`announcement\_seen`

- `announcement\_id` uuid fk`
- `user\_id` uuid fk`
- `seen\_at` timestamptz not null`
- `device\_type` text null` *(WEB/IOS/ANDROID)*
- pk `(announcement\_id, user\_id)`

`announcement\_ack`

- `announcement\_id` uuid fk`
- `user\_id` uuid fk`
- `ack\_at` timestamptz not null`
- `comment` text null`
- `source` text null` *(WEB/IOS/ANDROID)*
- pk `(announcement\_id, user\_id)`

`announcement\_likes` *(optional, if you keep Like)*

- `announcement\_id` uuid fk`
- `user\_id` uuid fk`
- `created\_at` timestamptz default now()`
- pk `(announcement\_id, user\_id)`

`announcement\_comments`

- `id` uuid pk`
 - `announcement\_id` uuid fk`
 - `user\_id` uuid fk`
 - `kind` text not null default 'COMMENT'` *(COMMENT/QUESTION)*
 - `body` text not null`
 - `status` text not null default 'VISIBLE'` *(VISIBLE/PENDING/HIDDEN/DELETED)*
 - `created\_at` timestamptz default now()`
 - `updated\_at` timestamptz default now()`
- **Indexes****
- `(announcement\_id, created\_at)`

1.6 Approvals / Workflow

`approval\_workflows`

- `id` uuid pk`
- `community\_id` uuid fk`
- `name` text not null`
- `is\_active` bool not null default true`

`approval\_steps`

- `id` uuid pk`
 - `workflow\_id` uuid fk`
 - `step\_order` int not null`
 - `role\_id` uuid fk -> roles(id)` *(who can approve this step)*
 - `required\_count` int not null default 1`
- **Indexes****
- unique `(workflow\_id, step\_order)`

`announcement\_approvals`

- `id` uuid pk`
- `announcement\_id` uuid fk`
- `step\_id` uuid fk`
- `decided\_by` uuid fk -> users(id)`
- `decision` text not null` *(APPROVED/REJECTED)*
- `comment` text null`
- `decided\_at` timestampz default now()`

1.7 Notifications / Reminders / Exports / Audit

`notification\_jobs`

- `id` uuid pk`
- `community\_id` uuid fk`
- `announcement\_id` uuid fk`
- `kind` text not null` *(PUBLISH/SCHEDULED_PUBLISH/REMINDER)*
- `channels` jsonb not null` *(in_app/push/email/sms)*
- `payload` jsonb not null`
- `status` text not null default 'QUEUED'` *(QUEUED/RUNNING/DONE/FAILED)*
- `created\_at` timestampz default now()`

`notification\_deliveries`

- `id` uuid pk`
- `job\_id` uuid fk`
- `user\_id` uuid fk`
- `channel` text not null`
- `status` text not null` *(SENT/DELIVERED/FAILED)*

- `provider\_msg\_id` text null`
- `created\_at` timestampz default now()`

`export\_jobs`

- `id` uuid pk`
- `community\_id` uuid fk`
- `requested\_by` uuid fk -> users(id)`
- `kind` text not null` *(ANALYTICS_DASHBOARD/NOTICE_REPORT)*
- `format` text not null` *(PDF/XLSX/CSV)*
- `filters` jsonb not null`
- `status` text not null default 'QUEUED`
- `file\_storage\_key` text null`
- `created\_at` timestampz default now()`

`audit\_logs`

- `id` uuid pk`
- `community\_id` uuid fk`
- `actor\_user\_id` uuid fk -> users(id)`
- `action` text not null` *(CREATE_NOTICE/PUBLISH/APPROVE/EXPORT/SEND_REMINDER/etc.)*
- `entity\_type` text not null` *(ANNOUNCEMENT/COMMENT/ATTACHMENT/etc.)*
- `entity\_id` uuid not null`
- `meta` jsonb not null default '{}`
- `created\_at` timestampz default now()`

****Indexes****

- `(community\_id, created\_at desc)`
- `(entity\_type, entity\_id)`

OpenAPI Spec (Starter) — `openapi.yaml` Skeleton

Below is a compact but valid starting contract. It covers ****all screens****: create/manage, detail, audience preview, attachments signed URL, approvals, analytics dashboard, drill-down report, exports, reminders, audit logs.

openapi: 3.0.3

info:

title: Digital Notice Board API

version: 1.0.0

servers:

- url: https://api.example.com

security:

- bearerAuth: []

components:

securitySchemes:

bearerAuth:

type: http

scheme: bearer

bearerFormat: JWT

schemas:

AnnouncementType:

type: string

enum: [GENERAL, TARGETED, RESTRICTED, EMERGENCY]

Priority:

type: string

enum: [NORMAL, HIGH, URGENT, EMERGENCY]

AnnouncementStatus:

type: string

enum: [DRAFT, PENDING_APPROVAL, APPROVED, SCHEDULED, PUBLISHED, EXPIRED, ARCHIVED, DELETED]

Announcement:

type: object

properties:

id: { type: string, format: uuid }
community_id: { type: string, format: uuid }
type: { \$ref: "#/components/schemas/AnnouncementType" }
priority: { \$ref: "#/components/schemas/Priority" }
category_id: { type: string, format: uuid, nullable: true }
title: { type: string }
body_format: { type: string, enum: [HTML, MD] }
body: { type: string }
ack_required: { type: boolean }
comments_enabled: { type: boolean }
pinned: { type: boolean }
highlight_banner: { type: boolean }
fullscreen_takeover: { type: boolean }
status: { \$ref: "#/components/schemas/AnnouncementStatus" }
starts_at: { type: string, format: date-time, nullable: true }
ends_at: { type: string, format: date-time, nullable: true }
created_at: { type: string, format: date-time }

TargetRules:

type: object
additionalProperties: true
description: JSON targeting rules (include/exclude).

paths:

/announcements:

get:

summary: List announcements (Manage Announcements)

parameters:

- in: query

name: status

schema: { type: string }

- in: query

name: q

schema: { type: string }

- in: query

name: cursor

schema: { type: string }

responses:

"200":

description: OK

post:

summary: Create announcement (draft)

requestBody:

required: true

content:

application/json:

schema:

type: object

required: [type, priority, title, body_format, body]

properties:

type: { \$ref: "#/components/schemas/AnnouncementType" }

priority: { \$ref: "#/components/schemas/Priority" }

category_id: { type: string, format: uuid, nullable: true }

title: { type: string }

body_format: { type: string, enum: [HTML, MD] }

body: { type: string }

ack_required: { type: boolean }

comments_enabled: { type: boolean }

target_rules: { \$ref: "#/components/schemas/TargetRules" }

responses:

"201": { description: Created }

/announcements/{id}:

get:

summary: Get announcement detail (RBAC enforced)

parameters:

- in: path

name: id

required: true

schema: { type: string, format: uuid }

responses:

"200": { description: OK }

"403": { description: Forbidden }

"404": { description: Not found }

patch:

summary: Update announcement (draft/scheduled rules apply)

parameters:

- in: path

name: id

required: true

schema: { type: string, format: uuid }

responses:

"200": { description: OK }

/announcements/{id}/publish:

post:

summary: Publish now or schedule publish

parameters:

- in: path

name: id
required: true
schema: { type: string, format: uuid }

requestBody:
required: true
content:
application/json:
schema:
type: object
properties:
mode: { type: string, enum: [NOW, SCHEDULE] }
starts_at: { type: string, format: date-time, nullable: true }
ends_at: { type: string, format: date-time, nullable: true }
channels:
type: object
properties:
in_app: { type: boolean }
push: { type: boolean }
email: { type: boolean }
sms: { type: boolean }

responses:
"200": { description: OK }

/announcements/targeting/preview:

post:
summary: Preview recipient count for targeting rules
requestBody:
required: true
content:
application/json:

schema:

type: object

required: [community_id, include]

properties:

community_id: { type: string, format: uuid }

include: { \$ref: "#/components/schemas/TargetRules" }

exclude: { \$ref: "#/components/schemas/TargetRules" }

mode: { type: string, enum: [SNAPSHOT, DYNAMIC] }

responses:

"200": { description: OK }

/announcements/{id}/seen:

post:

summary: Mark announcement as seen

parameters:

- in: path

name: id

required: true

schema: { type: string, format: uuid }

responses:

"204": { description: No Content }

/announcements/{id}/ack:

post:

summary: Acknowledge announcement

parameters:

- in: path

name: id

required: true

schema: { type: string, format: uuid }

requestBody:

content:

application/json:

schema:

type: object

properties:

comment: { type: string, nullable: true }

responses:

"204": { description: No Content }

/announcements/{id}/attachments:

get:

summary: List attachments

parameters:

- in: path

name: id

required: true

schema: { type: string, format: uuid }

responses:

"200": { description: OK }

/attachments/{attachmentId}/signed-url:

post:

summary: Get signed URL for preview/download

parameters:

- in: path

name: attachmentId

required: true

schema: { type: string, format: uuid }

requestBody:

required: true

content:

application/json:

schema:

type: object

required: [scope, announcement_id]

properties:

scope: { type: string, enum: [view, download] }

announcement_id: { type: string, format: uuid }

responses:

"200": { description: OK }

/announcements/{id}/analytics/summary:

get:

summary: Announcement analytics summary

parameters:

- in: path

name: id

required: true

schema: { type: string, format: uuid }

responses:

"200": { description: OK }

/announcements/{id}/analytics/recipients:

get:

summary: Drill-down recipients (seen/unseen/ack/unack)

parameters:

- in: path

name: id

required: true

schema: { type: string, format: uuid }

- in: query

name: status

schema: { type: string, enum: [seen, unseen, ack, unack] }

- in: query

name: cursor

schema: { type: string }

responses:

"200": { description: OK }

/announcements/{id}/reminders:

post:

summary: Send reminder batch

parameters:

- in: path

name: id

required: true

schema: { type: string, format: uuid }

responses:

"202": { description: Accepted }

/noticeboard/analytics/summary:

get:

summary: Dashboard KPIs

responses:

"200": { description: OK }

/noticeboard/analytics/announcements:

get:

summary: Dashboard table rows

responses:

"200": { description: OK }

/noticeboard/analytics/export:

post:

summary: Export dashboard/report (async)

responses:

"202": { description: Accepted }

/audit-logs:

get:

summary: Security & audit logs

responses:

"200": { description: OK }

Backlog (Epics → Stories) Mapped to Screens

Epic A — Foundations (RBAC + Tenant + Audit)

- A1: Community + user context (community selector, timezone, language)
- A2: RBAC permissions matrix (create/approve/publish/archive/analytics/audit)
- A3: Audit logging middleware (append-only)

Epic B — Manage Announcements (Web W1)

- B1: Announcements list + filters + search + sorting
- B2: Status lifecycle + transitions (draft/scheduled/published/expired/archived)
- B3: Row actions (duplicate/archive/delete/reminder)

Epic C — Create/Edit Announcement (Web W2 + W3)

- C1: Rich editor + CTA builder + validation
- C2: Attachments upload + metadata + per-attachment permissions
- C3: Scheduling/expiry/recurrence + auto-archive
- C4: Emergency flags (sticky/fullscreen/quiet hours override)

Epic D — Audience Targeting (Selector Panel)

- D1: Targeting UI (blocks/floors/units/roles/groups/users)
- D2: Recipient preview API + count breakdown
- D3: Recipient snapshot generation on publish

Epic E — Reader Experience (Mobile M1 + Web/Mobile Detail)

- E1: Mobile notices feed with tabs + pagination
- E2: Announcement detail (content, language variants, CTAs)
- E3: Seen tracking (threshold logic) + idempotent events
- E4: Acknowledgement flow (mandatory for critical)

Epic F — Attachments Viewer + Security (Attachment Viewer)

- F1: Signed URL service (view vs download)
- F2: PDF/image/video preview (web+mobile)

- F3: Watermark-on-download pipeline + auditing of downloads

Epic G — Engagement + Moderation (Web W4 + detail comments)

- G1: Comments/Q&A + permissions + locking
- G2: Moderation (hide/delete, pre/post moderation modes)
- G3: Feedback stats list

Epic H — Approval Workflow (Web W5/W6)

- H1: Workflow config + steps + required approvals
- H2: Approval queue + approve/reject + audit
- H3: Publish gates + emergency bypass policy

Epic I — Analytics (Web Dashboard + Drill-down + Exports)

- I1: Analytics dashboard KPIs + charts + table
- I2: Per-notice drill-down recipients + filtering
- I3: Exports (PDF/XLSX async jobs)
- I4: Reminders (eligibility + rate limits + delivery logging)

Epic J — Digital Display Integration (Web W7)

- J1: Display devices + playlists + rotation rules
- J2: Sync pipeline + emergency override
- J3: Device delivery logs

Epic K — Security & Audit Logs (Web W8)

- K1: Audit log UI + filters + export
- K2: Policy: retention, soft-delete visibility

